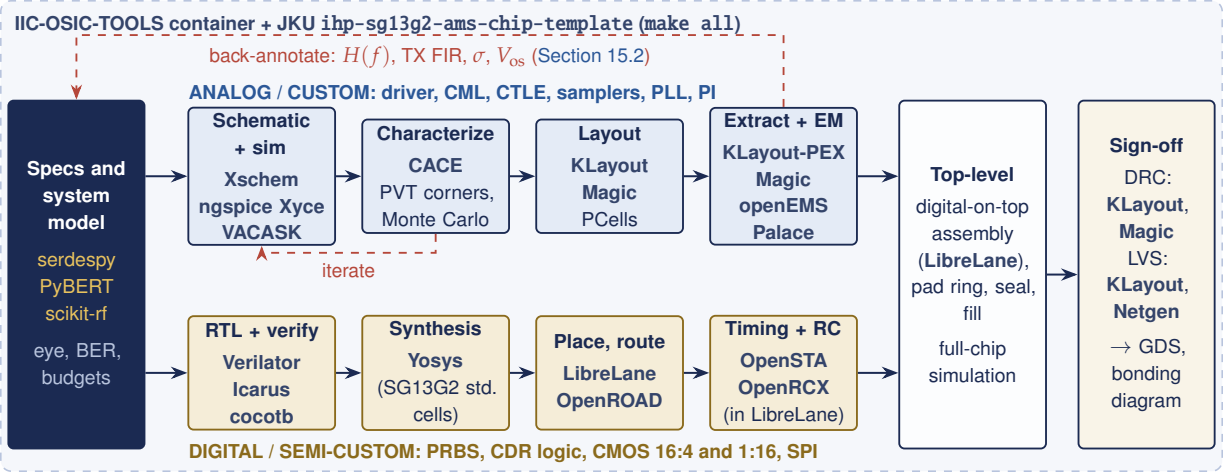

IIC-OSIC-TOOLS container + JKU ihp-sg13g2-ams-chip-template (make all)
back-annotate: $H(f)$, TX FIR, σ, V_{os} (Section 15.2)
ANALOG / CUSTOM: driver, CML, CTLE, samplers, PLL, PI
Specs and system model
serdespy
PyBERT
scikit-rf
eye, BER, budgets
Schematic + sim
Xschem
ngspice Xyce
VACASK
Characterize
CACE
PVT corners, Monte Carlo
iterate
Layout
KLayout
Magic
PCells
Extract + EM
KLayout-PEX
Magic
openEMS
Palace
Top-level
digital-on-top assembly (LibreLane), pad ring, seal, fill
full-chip simulation
Sign-off
DRC: KLayout, Magic
LVS: KLayout, Netgen
→ GDS, bonding diagram
RTL + verify
Verilator
Icarus
cocotb
Synthesis
Yosys
(SG13G2 std. cells)
Place, route
LibreLane
OpenROAD
Timing + RC
OpenSTA
OpenRCX
(in LibreLane)
DIGITAL / SEMI-CUSTOM: PRBS, CDR logic, CMOS 16:4 and 1:16, SPI